ASHWINI KUMARI

Profile

Technical Lead with 9+ years of experience in Automotive Embedded Software Development across Powertrain, EPS, EMS, ADAS, and EV. Proven expertise in AUTOSAR, CAN, Diagnostics (UDS), Model Based Development and Vehicle level testing. Strong track record in **Leading** a **Team**, customer interactions, software releases, and end-to-end ECU lifecycle delivery for OEMs such as TATA, Mahindra, Volvo.

PROFESSIONAL EXPERIENCE

05/2022 – Present | **Technical Lead** | Bengaluru

HL Mando Softtech India Pvt. Ltd.

• Leading end-to-end EPS software development for Passenger and Commercial Vehicles (TATA, Volvo, Mahindra).
• Direct interaction with OEMs for requirements and any other Open Points or Issues, done requirements analysis, task allocation, code reviews and SW delivery to OEM.
• Developed AUTOSAR SWCs, runnables, and communication stacks using EBTresos.
• Implemented ADAS features, CAN communication and Diagnostics.
• Performed static code analysis and ensured MISRA C compliance using Polyspace.
• Handled defect tracking, sprint planning, and release process using JIRA and PTC.

11/2021 – 04/2022 | **Assistant Manager** | Gurugram

Hero Electric Vehicle Pvt. Ltd.

• Worked on Electric Vehicle system testing, motor calibration, and BMS validation.
• Coordinated requirement clarification and fixes with BMS suppliers.

08/2016 – 11/2021 | **Senior Software Engineer** | Bengaluru

Robert Bosch

• Developed and tested EMS and Powertrain ECU SW for Global OEMs.
• Direct interaction with OEMs for requirements and any other Open Points or Issues, Handled requirement analysis, design, coding, integration and SW delivery.
• Implemented CAN communication and UDS Diagnostics.

• Performed HIL testing using dSPACE LabCar and vehicle level validation.

SKILLS

Core Skills :
• **Automotive Powertrain Domain**, **Embedded C, Model Based Development.**
• **AUTOSAR** Classic, COM, SWC, BSW, RTE, Rhapsody, EBTresos.
• **CAN**, CAN-FD, LIN overview, **UDS** (ISO 14229), CAPL Scripting, Vector DaVinci.
• **Diagnostics, ADAS**, Cluster, EPS, **Engine Management System**, BMS.
• **MATLAB, Simulink, ASCET**, INCA, CANoe, CANalyzer, CANape, Vector Tools.
• HIL/MIL Testing, LabCar, TPT, VTestStudio, Debugging Tools(Trace32 and UDE), Unit Testing.
• MISRA C, **Polyspace**, Static Code Analysis, FMEA overview and ASIL classification with **ISO 26262**, **RTOS** Concept.
• JIRA, **PTC** Integrity, **Windchill**, Jenkins.
• Infineon TriCore Microcontroller.
• Development Models: SDLC, V-Model, Waterfall, Agile, ASPICE Overview.
• Electric Vehicles, CNG Vehicles, Hybrid Vehicles and Gasoline Vehicles.

KEY PROJECTS

BEV, Z121, S240, S220 SWs
Customer : Mahindra Vehicles
Details :
- Leading a team of 7 members.
- Following and using **SDLC** and V-Model with end to delivery of the SW.
- Involved in Requirement Gathering and Analysis with **ASPICE** Process.
- CAN Message implementation.
- Read and Write DID implementation.
- ADAS feature implementation.
- **Autosar** (SWC, RTE, COM, BSW, MCAL) and Embedded C based implementation.
- Usage of development tools like Vector Geny, Da-Vinci, Tresos.
- Software Integration testing, Unit testing using test Automotive test automation framework as well as hardware.
- Use of Vector Tools like Canoe, Davinci and writing the CAPL script for testing.
- Writing the test cases using python script in VTestStudio for testing SW.
- Use of Jenkins for compilation and build package creation.
- Debugging the SW using UDE, TRACE32 AND JTAG.

Harrier, Safari, Nexon(ADAS, EV, Diesel, Petro, Bifuel, CNG), Osprey, Tiago, Tigor, Altroz SWs
Customer : TATA Motors
Details :
- Leading a team of 7 members.
- Following and using **SDLC** and V-Model with end to delivery of the SW.
- Involved in Requirement Gathering and Analysis with **ASPICE** Process.
- CAN Message implementation.
- Read and Write DID implementation.
- ADAS feature implementation.
- **Autosar** (SWC, RTE, COM, BSW) based implementation.

- Usage of development tools like Vector Geny, Da-Vinci, Tresos.
- Software Integration testing, Unit testing using test Automotive test automation framework as well as hardware.
- Use of Vector Tools like Canoe and writing the CAPL script.
- Use of Jenkins for compilation and build package creation.

Volvo Eicher Titan
Customer : Volvo Trucks
Details :
- Leading a team of 7 members.
- Following and using **SDLC** and V-Model with end to delivery of the SW.
- Involved in Requirement Gathering and Analysis with **ASPICE** Process.
- CAN Message implementation.
- Read and Write DID implementation.
- ADAS feature implementation.
- **Autosar** (SWC, RTE, COM, BSW) based implementation.
- Usage of development tools like Vector Geny, Da-Vinci, Tresos.
- Software Integration testing, Unit testing using test Automotive test automation framework as well as hardware.
- Use of Vector Tools like Canoe and writing the CAPL script.
- Use of Jenkins for compilation and build package creation.

Optima Electric 2 Wheeler Vehicle
Customer : Hero Electric
- **Details :** Involved in Requirement fix, Requirement sharing to Supplier for implementation, CAN message implementation and Calibration of the SW.

EDUCATION

05/2012 – 05/2016	**B.Tech** *Kalinga Institute of Industrial Technology* Percentage : 81%	Bhubaneswar, Orissa

Languages

English — Fluent

Hindi — Fluent